

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

Give Away Platform

Version: 1.1

Architecture: Microservices (4 Services)

Prepared By: Product & Engineering Team

Project Type: Charity, NGO & Donation Management Platform

1. PROJECT OVERVIEW

Purpose

Give Away is a digital donation and charity management platform operated by AJA Abhayahastam. Donors contribute money or items directly to AJA Abhayahastam, which collects, verifies, and manages all donations. Registered NGOs and individual receivers can then request assistance from AJA, subject to document verification and admin approval. The platform ensures transparency and accountability at every step.

The platform supports:

- NGO Registration & Verification
- Receiver Financial Assistance Requests
- Donor Money Donations
- Donor Item Donations
- NGO Item & Fund Requests
- Inventory & Fund Ledger Management
- Donation Utilization Tracking
- Notifications

2. USER ROLES

Super Admin

Responsibilities

- Verify NGOs and Receivers
- Manage users and accounts
- Approve/reject assistance and NGO requests
- Manage funds, inventory, and ledgers
- View reports and audit logs
- Manage platform settings

Donor

Responsibilities

- Register/Login
- Donate money
- Donate items

- Track donation status
- View utilization / impact updates

Receiver

Responsibilities

- Register/Login
- Complete KYC / document verification
- Apply for financial assistance only
- Upload supporting documents (e.g. medical)
- Track application status

NGO Partner

Responsibilities

- Register
- Upload verification documents
- Wait for admin approval
- Request items and/or funds
- Manage beneficiaries
- Maintain/view inventory allocated to them
- View reports

Guest

Responsibilities

- Browse public landing pages
- View verified NGO directory / impact summaries
- Access registration and login entry points

3. SYSTEM ARCHITECTURE

React Website

React Native Mobile App

Admin Portal

|
v

API Gateway

|
|-----|
| | | |

Identity Profile & Core Notification

Service Verification Charity Service

Service Service

|-----|
|

PostgreSQL (one DB per service)

|

AWS S3 / MinIO (documents & media)

Note: Logical domains from earlier 6-service design are consolidated into 4 deployable FastAPI services for simpler delivery.

3.1 TECHNOLOGY STACK

- Web frontend: React
- Mobile frontend: React Native
- Admin portal: React
- Backend: FastAPI microservices
- API: REST via API Gateway
- Auth: JWT + refresh tokens
- DB: PostgreSQL (one DB per service)
- Files: AWS S3 / MinIO
- Notifications: In-app, Email, SMS, Push

4. BUSINESS FLOW

NGO Registration & Verification

Register

↓

Account Created (REGISTERED)

↓

Upload Documents

↓

DOCUMENTS_SUBMITTED

↓

Admin Review (UNDER_REVIEW)

↓

APPROVED / REJECTED

↓

If Approved: VERIFIED & Modules Unlocked

Receiver Financial Assistance

Register

↓

Verify Identity / Documents

↓

Submit Financial Assistance Application

↓

SUBMITTED

↓

Admin Review

↓

APPROVED / REJECTED

↓

If Approved: Fund Allocation / Disbursement

↓

COMPLETED

↓

Notify Receiver

Donor Money Donation

Login

↓

Choose Fund / Program / Pool

↓

Enter Amount

↓

Payment Gateway

↓

Payment Success

↓

Ledger Entry

↓

Utilization Update / Receipt

↓

Notify Donor

Donor Item Donation

Login

↓

List Item (Category, Details, Quantity, Photos, Pickup)

↓

PENDING_ASSIGNMENT

↓

Admin Assigns

↓

PICKUP_SCHEDULED

↓

Picked Up

↓

Received Into Inventory



Inventory Updated



Notify Donor

NGO Request Flow

Verified NGO Submits Item or Fund Request



Admin Review



Approve / Reject



If Approved: Allocate Inventory or Disburse Funds



Update Ledger / Inventory



Notify NGO

5. SERVICE 1: IDENTITY SERVICE

Purpose

Manage authentication, user accounts, roles, and login security across the platform.

Modules

Authentication

- Login
- Logout
- JWT Token
- Password Reset
- OTP Verification

User Account Management

- Create Account
- Update Account
- Deactivate Account
- Role Assignment

Role Management

- Define Roles
- Assign Roles
- Update Roles

Database: identity_db

Table 1: roles

Field	Type
role_id	BIGINT
role_name	VARCHAR
description	VARCHAR

Table 2: users

Field	Type
user_id	BIGINT
role_id	BIGINT
full_name	VARCHAR
email	VARCHAR
mobile	VARCHAR
password_hash	VARCHAR
status	VARCHAR
created_at	TIMESTAMP

Table 3: refresh_tokens

Field	Type
token_id	BIGINT
user_id	BIGINT
token	VARCHAR
expires_at	TIMESTAMP
created_at	TIMESTAMP

Table 4: login_audit

Field	Type
audit_id	BIGINT
user_id	BIGINT
login_time	TIMESTAMP
ip_address	VARCHAR
status	VARCHAR

Table 5: otp_verifications

Field	Type
otp_id	BIGINT
user_id	BIGINT
otp_code	VARCHAR

purpose	VARCHAR
expires_at	TIMESTAMP
verified_status	VARCHAR

Total Tables = 5

6. SERVICE 2: PROFILE & VERIFICATION SERVICE

Purpose

Manage donor, receiver, and NGO profiles, beneficiaries, programs, addresses, and document verification for NGOs, Receivers, and Donors, including the admin review queue.

Modules

Donor Profile

- Create Profile
- Update Profile
- View Donation History

Receiver Profile

- Create Profile
- Update Profile
- View Application History

NGO Organization

- Create NGO Profile
- Update NGO Profile
- View Verification Status

Beneficiary Management

- Add Beneficiary
- Update Beneficiary
- View Beneficiaries

Programs Catalog

- Create Program
- Update Program
- Activate/Deactivate Program

Admin User Management

- View Users
- Update User Status
- Assign Roles

NGO Verification

- Submit Documents
- Track Status

Receiver Verification

- Submit Documents
- Track Status

Donor Verification (if applicable)

- Submit Documents
- Track Status

Admin Verification Queue

- Review Requests
- Approve/Reject
- Add Rejection Reason

Document Storage References

- Store Document URL
- Link to Verification Request

Database: profile_verification_db

Table 1: donor_profiles

Field	Type
donor_id	BIGINT
user_id	BIGINT
full_name	VARCHAR
mobile	VARCHAR
email	VARCHAR
address_id	BIGINT

Table 2: receiver_profiles

Field	Type
receiver_id	BIGINT
user_id	BIGINT
full_name	VARCHAR
mobile	VARCHAR
email	VARCHAR
address_id	BIGINT
verification_status	VARCHAR

Table 3: ngo_profiles

Field	Type
ngo_id	BIGINT
user_id	BIGINT
ngo_name	VARCHAR
registration_number	VARCHAR
contact_person	VARCHAR
mobile	VARCHAR

address_id	BIGINT
verification_status	VARCHAR

Table 4: beneficiaries

Field	Type
beneficiary_id	BIGINT
ngo_id	BIGINT
name	VARCHAR
age	INT
details	TEXT

Table 5: programs

Field	Type
program_id	BIGINT
program_name	VARCHAR
description	TEXT
category	VARCHAR
status	VARCHAR

Table 6: addresses

Field	Type
address_id	BIGINT
line1	TEXT
city	VARCHAR
state	VARCHAR
pincode	VARCHAR

Table 7: verification_requests

Field	Type
request_id	BIGINT
user_id	BIGINT
request_type	VARCHAR
status	VARCHAR
submitted_at	TIMESTAMP

Request Types:

- NGO
- RECEIVER
- DONOR

Table 8: verification_documents

Field	Type
document_id	BIGINT
request_id	BIGINT
document_type	VARCHAR
file_url	VARCHAR
uploaded_at	TIMESTAMP

Table 9: verification_status_history

Field	Type
history_id	BIGINT
request_id	BIGINT
status	VARCHAR
changed_at	TIMESTAMP

Table 10: rejection_reasons

Field	Type
reason_id	BIGINT
request_id	BIGINT
reason	TEXT
created_at	TIMESTAMP

Total Tables = 10

7. SERVICE 3: CORE CHARITY SERVICE

Purpose

Manage money donations, item donations, receiver assistance applications, NGO item/fund requests, inventory, pickup scheduling, fund ledger, and allocation/disbursement operations.

Modules

Money Donations

- Donate to Fund/Program
- Payment Tracking
- Utilization Updates

Item Donations

- List Item
- Track Pickup Status
- Track Inventory Status

Receiver Assistance Applications

- Submit Application
- Upload Documents
- Track Status

NGO Item Requests

- Request Items
- Track Status

NGO Fund Requests

- Request Funds
- Track Status

Status Tracking

- Donation Status History
- Application Status History

Inventory Management

- Add Item
- Update Item
- Track Stock

Pickup Scheduling

- Schedule Pickup
- Update Pickup Status

Fund Ledger

- Record Transaction
- View Ledger

Allocation/Disbursement

- Allocate Inventory
- Disburse Funds

Reports Support Data

- Aggregate Inventory Data
- Aggregate Fund Data

Database: charity_db

Table 1: money_donations

Field	Type
donation_id	BIGINT
donor_id	BIGINT
program_id	BIGINT
amount	DECIMAL
payment_status	VARCHAR
donated_at	TIMESTAMP

Table 2: item_donations

Field	Type
item_donation_id	BIGINT
donor_id	BIGINT

category	VARCHAR
description	TEXT
quantity	INT
pickup_address_id	BIGINT
status	VARCHAR

Table 3: assistance_applications

Field	Type
application_id	BIGINT
receiver_id	BIGINT
purpose	VARCHAR
amount_requested	DECIMAL
status	VARCHAR
submitted_at	TIMESTAMP

Table 4: ngo_item_requests

Field	Type
request_id	BIGINT
ngo_id	BIGINT
item_category	VARCHAR
quantity_requested	INT
status	VARCHAR

Table 5: ngo_fund_requests

Field	Type
request_id	BIGINT
ngo_id	BIGINT
amount_requested	DECIMAL
purpose	TEXT
status	VARCHAR

Table 6: donation_status_history

Field	Type
history_id	BIGINT
donation_id	BIGINT
donation_type	VARCHAR
status	VARCHAR
changed_at	TIMESTAMP

Table 7: application_status_history

Field	Type
history_id	BIGINT
application_id	BIGINT
status	VARCHAR
changed_at	TIMESTAMP

Table 8: inventory_items

Field	Type
item_id	BIGINT
category	VARCHAR
description	TEXT
quantity	INT
status	VARCHAR

Table 9: inventory_transactions

Field	Type
transaction_id	BIGINT
item_id	BIGINT
transaction_type	VARCHAR
quantity	INT
transaction_date	TIMESTAMP

Table 10: pickup_schedules

Field	Type
pickup_id	BIGINT
item_donation_id	BIGINT
pickup_date	DATE
pickup_time	TIME
status	VARCHAR

Table 11: fund_pools

Field	Type
pool_id	BIGINT
pool_name	VARCHAR
balance	DECIMAL

Table 12: fund_ledger

Field	Type
-------	------

ledger_id	BIGINT
pool_id	BIGINT
transaction_type	VARCHAR
amount	DECIMAL
transaction_date	TIMESTAMP

Table 13: disbursements

Field	Type
disbursement_id	BIGINT
application_id	BIGINT
amount	DECIMAL
disbursed_at	TIMESTAMP
status	VARCHAR

Table 14: allocations

Field	Type
allocation_id	BIGINT
ngo_request_id	BIGINT
item_id	BIGINT
quantity_allocated	INT
allocated_at	TIMESTAMP

Total Tables = 14

8. SERVICE 4: NOTIFICATION SERVICE

Purpose

Manage in-app, email, SMS, and push notifications across the platform.

Modules

In-App

- Send Notification
- Mark as Read

Email

- Send Email
- Track Delivery

SMS

- Send SMS
- Track Delivery

Push

- Send Push
- Track Delivery

Templates

- Create Template
- Update Template

Delivery Logs

- Log Delivery Status
- View Logs

Preferences

- Set Channel Preferences
- View Preferences

Database: notification_db

Table 1: notifications

Field	Type
notification_id	BIGINT
user_id	BIGINT
title	VARCHAR
message	TEXT
status	VARCHAR
created_at	TIMESTAMP

Table 2: notification_templates

Field	Type
template_id	BIGINT
template_name	VARCHAR
channel	VARCHAR
content	TEXT

Table 3: notification_delivery_logs

Field	Type
log_id	BIGINT
notification_id	BIGINT
channel	VARCHAR
delivery_status	VARCHAR
delivered_at	TIMESTAMP

Table 4: user_notification_preferences

Field	Type
preference_id	BIGINT
user_id	BIGINT

channel	VARCHAR
enabled	BOOLEAN

Total Tables = 4

9. VERIFICATION STATUS FLOW

REGISTERED

↓

DOCUMENTS_SUBMITTED

↓

UNDER_REVIEW

↓

VERIFIED / REJECTED / SUSPENDED

10. DONATION STATUS FLOW

Money Donation

INITIATED

↓

PAYMENT_PENDING

↓

CONFIRMED

↓

UTILIZED

Item Donation

LISTED

↓

PENDING_ASSIGNMENT

↓

PICKUP_SCHEDULED

↓

PICKED_UP

↓

RECEIVED

↓

IN_INVENTORY

↓

ALLOCATED / DISTRIBUTED

11. ASSISTANCE STATUS FLOW

DRAFT

↓

SUBMITTED

↓

UNDER_REVIEW

↓

APPROVED / REJECTED

↓

DISBURSED

↓

COMPLETED

12. NGO REQUEST STATUS FLOW

SUBMITTED

↓

UNDER_REVIEW

↓

APPROVED / REJECTED

↓

ALLOCATED / DISBURSED

↓

COMPLETED

13. FRONTEND APPLICATIONS

React Website

Pages

- Home
- About
- Contact
- Success Stories
- Login
- Register

Donor Module

- Login
- Home/Dashboard
- Donate Money
- Donate Item
- My Donations
- My Impact
- Profile
- Settings
- Notifications

Receiver Module

- Login
- Home/Dashboard
- Apply Assistance
- My Applications
- Profile
- Settings
- Notifications

NGO Module

- Login
- Home/Dashboard
- Verification

- Request Donations
- Request Funds
- Inventory
- Beneficiaries
- Reports
- Profile
- Settings
- Notifications

React Native Mobile App

Donor Pages

- Login
- Home/Dashboard
- Donate Money
- Donate Item
- My Donations
- My Impact
- Profile
- Settings
- Notifications

Receiver Pages

- Login
- Home/Dashboard
- Apply Assistance
- My Applications
- Profile
- Settings
- Notifications

NGO Pages

- Login
- Home/Dashboard
- Verification
- Request Donations
- Request Funds
- Inventory
- Beneficiaries
- Profile
- Settings
- Notifications

Admin Portal

Pages

- Dashboard
- Verification Queue
- Users

- NGOs
- Donations
- Funds
- Inventory
- Reports
- Settings
- Notifications

14. FINAL SUMMARY

Component	Count
Microservices	4
Databases	4
Tables	33
User Roles	5
Frontend Apps	3